MIL–P–12953A
3 MARCH 1955
SUPERSEDING
MIL–P–12953(SigC)
6 AUGUST 1953

MILITARY SPECIFICATION

# PUBLIC ADDRESS SET (REPRODUCING EQUIPMENT MC-364-( )) AND PUBLIC ADDRESS SET AN/UIH-2( )

*This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy, and the Air Force.*

## 1. SCOPE

1.1 This specification covers two types of equipments designated as Reproducing Equipment MC–364–( ) which operates at two selectable speeds of 33⅓ and 78.26 r.p.m. and Public Address Set AN/UIH–2( ) which operates at three selectable speeds of 33⅓, 45, and 78.26 r.p.m. (See 6.1 and 6.5.)

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

### SPECIFICATIONS

FEDERAL

NN–B–621 —Boxes; Wood, Nailed and Lock Corner.
UU–T–111 —Tape, Paper, Gummed (Sealing and Securing).
PPP–B–601 —Boxes; Wood-Cleated-Plywood.

MILITARY

MIL–P–75 —Packaging, Packing and Container-Marking of Electron Tubes.
JAN–P–106 —Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN–P–108 —Packaging and Packing for Overseas Shipment—Boxes; Fiberboard (V-Board and W-Board), Exterior, and Interior.
MIL–P–116 —Preservation, Methods of.
JAN–P–117 —Packaging and Packing for Overseas Shipment—Bags, Interior Packing.
JAN–P–127 —Packaging and Packing for Overseas Shipment—Tape, Adhesive, Pressure-Sensitive, Water Resistant.
JAN–T–152 —Treatment, Moisture— and Fungus-Resistant of Communications Electronic, and Associated Electrical Equipment. General Process for.
MIL–B–131 —Barrier Material; Water Vapor-Proof, Flexible.
MIL–C–3885 —Cable Assemblies and Cord Assemblies, Electrical (For Use in Electronic, Communication, and Associated Electrical Equipment).
MIL–M–11101 —Microphones, Dynamic.
MIL–P–11268 —Parts, Materials, and Processes, Used in Signal Corps Equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:53Z
Check the source to verify that this is the current version before use.

MIL-S-11748 —Suppression Requirements for Electrical and Electronic Equipment (Except Internal Combustion Engine-Driven Equipment.)
MIL-L-13078 —Loudspeaker, Dynamic (LS-112 ( )/U, LS-116( )/U, LS-148( )/U, LS-160( )/U, LS-179( )/U, LS-181( )/U) and Loudspeaker Assembly (LS-154( )-/U and LS-206( )/U).

U. S. ARMY

19-10 —Filler, Interior Packing, Fiberboard, Corrugated, Single and Double faced, For General Packing Purposes.

## STANDARDS

MILITARY

MIL-STD-15 —Electrical and Electronic Symbols.
MIL-STD-105 —Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129 —Marking for Shipment and Storage.
MIL-STD-130 —Identification Marking for U. S. Military Property.

## DRAWINGS

SIGNAL CORPS

SC-D-15800 —Oscilloscope BC-000-( ), Circuit Label (Schematic Diagram).
SC-D-15801 —Junction Box JB-00-( ), Circuit, Label, Wiring Diagram.
SC-D-15914 —Moisture Resistance Test Cycle for Ground Signal Equipment.
SC-D-19999 —Standard Extreme-Temperature Cycle.
SC-D-20096 —Rectifier PP-0/0, Sample Circuit Label, Schematic and Wiring Diagram.
SC-D-25128 —Power Cable Assembly CX-1208/U.
SC-D-25129 —Special Purpose Cable Assembly CX-1207/U.
SC-D-25130 —Power Cable Assembly CX-1235/U.
SC-D-27204 —Turntable MX-932-( )/U Drawing and Data List.

(Copies of specifications, standards and drawings, required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 **Other publications.** — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification Ratings, Rules and Regulations.

(Copies of Uniform Freight Classification Ratings, Rules and Regulations may be obtained from Association of American Railroads, Freight Container Bureau, Chicago, Illinois.)

## 3. REQUIREMENTS

3.1 **Procurement model.**—The procurement model shall be understood as the one which, in accordance with the invitation for bids, will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein, the equipment shall conform to the following:

3.1.1 *Construction.* — Construction of the equipment shall conform to the procurement model.

3.1.2 *Functions.* — The equipment shall incorporate all functions provided by the procurement model.

3.1.3 *Performance.* — Performance of the equipment shall be at least equal to that of the procurement model. When any performance characteristic of the procurement model is not specified herein, the following shall apply: As soon as the procurement model is received by the contractor, he shall adjust it for optimum performance and prepare to



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:53Z
Check the source to verify that this is the current version before use.

conduct all measurements necessary to determine the performance characteristics which are not specified herein. The contracting officer shall be notified of the date when the measurements are to be taken, sufficiently in advance to enable his representative to witness them. Measurements shall be taken by the contractor and three record copies thereof furnished to the contracting officer.

3.2 **Preproduction samples.**

3.2.1 *Quantities.* — After award of contract, in addition to the samples listed in Specification MIL–P–11268, two complete equipments shall be furnished for preproduction inspection and approval.

3.2.2 *Compliance with specified requirements.*

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.2.2.3 for exception.)

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.2.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.2.3 When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.2.3 *Test data.* — The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.2.4 *Reference standards.* — After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3 **Parts, materials, and processes; general.**—In addition to any requirements of this specification, covering parts, materials, and processes such items shall conform to Specification MIL–P–11268 including the selection requirements therein.

3.4 **Reproducing Equipment MC–364–( ).** — Reproducing Equipment MC–364–( ) shall consist of Loudspeaker Assembly LS–148( )/U, Microphone, Dynamic M–23–( )/U, Special Purpose Cable Assembly CX–1207/U (25 ft., 0 in.), Power Cable Assembly CX–1208/U (25 ft., 0 in.), 2 Power Cable Assembly CX–1235/U (50 ft., 0 in.), and Turntable MX–932( )/U.

3.5 **Public Address Set AN/UIH–2( )**— Public Address Set AN/UIH–2( ) shall consist of Loudspeaker Assembly LS–148–( )/U, Microphone, Dynamic M–23( )/U, Special Purpose Cable Assembly CX–1207/U (25 ft., 0 in.), Power Cable Assembly CX–1208/U (25 ft., 0 in.), 2 each Power Cable Assembly CX–1235/U (50 ft., 0 in.) and Reproducer RP–104( )/UIH–2.

3.6 **Loudspeaker Assembly LS–148( )/U.** — Loudspeaker Assembly LS–148( )/U shall be constructed in accordance with Specification MIL–L–13078.

3.7 **Microphone, Dynamic, M–23( )/U.**— Microphone, Dynamic, M–23( )/U shall be constructed in accordance with the procurement model and shall meet the test requirements of microphone M–43/U in accordance



Source: https://assist.dla.mil -- Downloaded: 202640-17T23:53Z
Check the source to verify that this is the current version before use.

with Specification MIL-M-11101.

3.8 **Special Purpose Cable Assembly CX-1207/U.**—Special Purpose Cable Assembly CX-1207/U shall be constructed in accordance with Drawing SC-D-25129 and shall be provided for connection of the microphone to the amplifier.

3.9 **Power Cable Assembly CX-1208/U.**—Power Cable Assembly CX-1208/U shall be constructed in accordance with Drawing SC-D-25128 and shall be provided for connection of Turntable MX-932( )/U or Reproducer RP-104( )/UIH-2 to power source.

3.10 **Power Cable Assembly CX-1235/U.**—Power Cable Assembly CX-1235/U shall be constructed in accordance with Drawing SC-D-25130 and shall be furnished to connect each speaker during operation.

3.11 **Turntable MX-932( )/U.** — Turntable MX-932( )/U shall be constructed in accordance with Drawing SC-D-27204.

3.12 **Reproducer RP-104( )/UIH-2.** — Reproducer RP-104( )/UIH-2 shall be constructed in accordance with the procurement model.

3.13 **Operating voltages.** — Reproducing Equipment MC-364-( ) and Public Address Set AN/UIH-2( ) shall operate and maintain the required turntable speeds when powered from nominal 115/230 volt, 50 to 60 cycle alternating current supply sources.

3.14 **Power consumption.**—The maximum power consumption from primary source when operating the complete equipment, Reproducing Equipment MC-364-( ) or Public Address Set AN/UIH-2( ) shall be 100 watts.

3.15 **Power output and distortion.**—A normal power output of ten watts at 3 percent total harmonic distortion, and peak power output of 12 watts a 5 percent total harmonic distortion, or less, shall be obtained when tested in accordance with 4.5.

3.16 **Frequency response.**—When measured as specified in 4.6, the frequency response characteristic of the amplifier section of Reproducing Equipment MC-364-( ) or Public Address Set AN/UIH-2( ) shall be within the limits shown on figure 1.

3.17 **Hum level.**

3.17.1 *Hum level of microphone channel.*— Resultant hum level from the microphone channel at full amplifier gain shall be 28 db or less above 1 millivolt as measured across 3 ohms. When amplifier gain is gradually decreased to minimum, the hum level shall not be greater at any point than at full amplifier gain.

3.17.2 *Hum level of phono channel.* — Resultant hum level from the phono channel at full amplifier gain shall be 8 db or less above 1 millivolt as measured across 3 ohms. When amplifier gain is gradually decreased to minimum, the hum level shall not be greater at any point than at full amplifier gain.



Source: https://assist.dla.mil -- Downloaded: 202640-17T23:53Z
Check the source to verify that this is the current version before use.

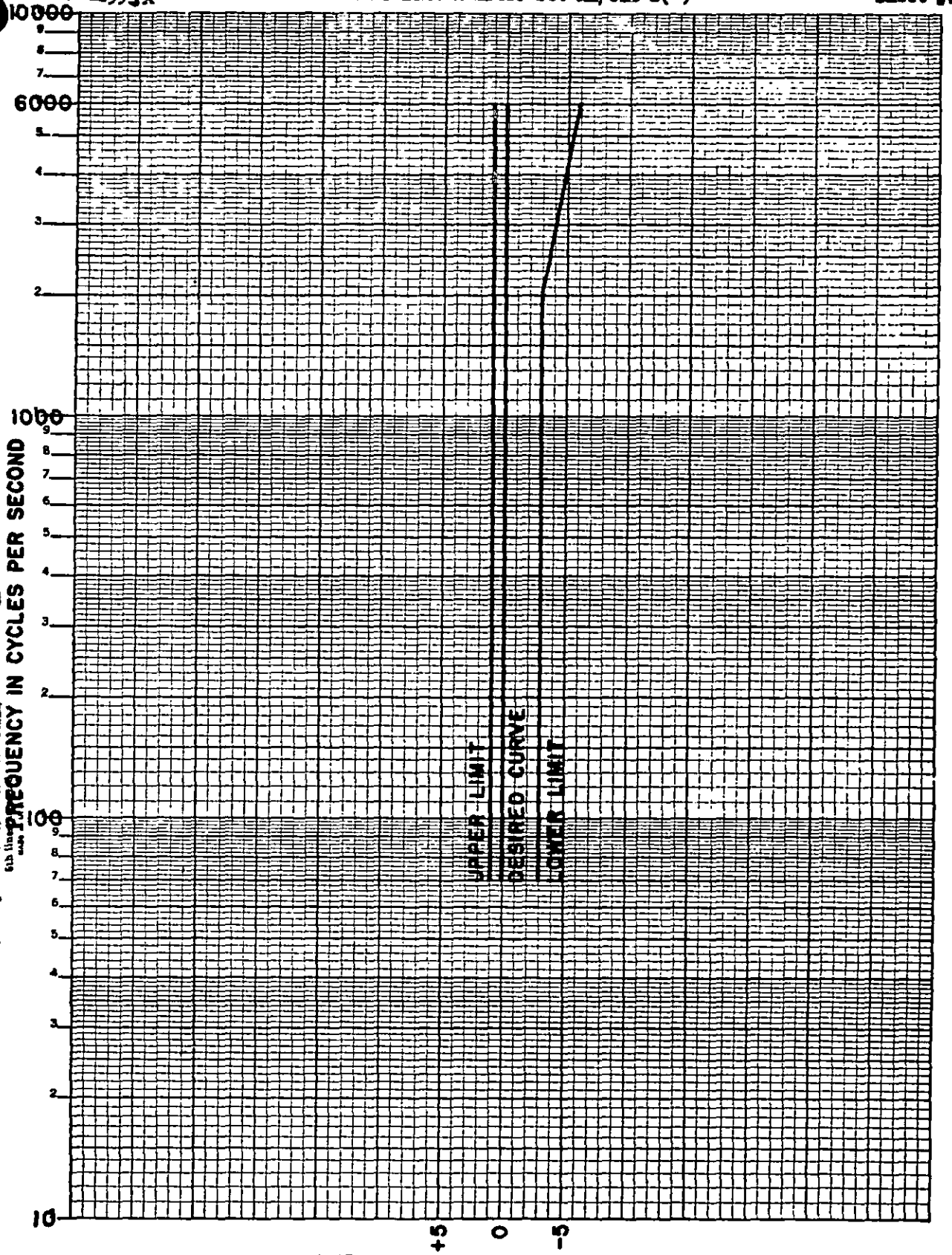



FIG. 1 FREQUENCY RESPONSE CHARACTERISTIC

Source: https://assist.dla.mil -- Downloaded: 2026-10-14 23:53Z
Check the source to verify that this is the current version before use.

3.18 **Turntable Speeds.**

3.18.1 The turntable for turntable MX-932 ( )/U shall operate at two selectable speeds of 33⅓ and 78.26 r.p.m. The turntable shall be capable of playing all types of commercial lateral type recordings of 10, 12, and 16 inch diameters.

3.18.2 The turntable for Reproducer RP-104( )/UIH-2 shall operate at three selectable speeds of 33⅓, 45, and 78.26 r.p.m. The turntable shall be capable of playing all types of commercial lateral type recordings of 7, 10, 12, and 16 inch diameters. Provision shall also be made for adapting the large size hole in the 45 r.p.m. records to mount on the standard diameter turntable motor shaft.

3.18.3 Selection of desired operation shall be readily accomplished in a convenient manner.

3.18.4 The turntable shall be positively indexed at any desired speed of operation and shall contain a simple adjustment to obtain the specified speeds when operating the equipment from primary power source of any frequency within the 50 to 60 cycle range and any voltage within plus or minus 10 percent of those specified in 3.13.

3.19 **Needle pressure.**—The pressure of the needle in the pickup shall be 15 to 20 grams.

3.20 **Radio interference suppression.**—The equipment shall conform to the requirements of Specification MIL-S-11748 and the interference limits set forth therein, with the antenna of the interference measuring equipment located 10 feet from the equipment under test.

3.21 **Tropicalization.** — The assembled equipment shall be treated in accordance with Specification JAN-T-152.

3.22 **Interchangeability.** — Corresponding components and replaceable subassemblies and parts on contract shall be physically and functionally interchangeable as units without modification thereof or of other items with which the units are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. Whether subassemblies and parts are to be considered replaceable shall be as determined by the contracting officer.

3.23 **Service conditions.**—The equipment shall meet the following service conditions:

3.23.1 *Equipment operating.*—The equipment, after a warmup period of ½ hour, shall meet the requirements of this specification under any probable combination of the following:

(a) *Operation.* — Continuous operation in any specified manner, for a period of at least 1 year, alternating 8 hours on and 16 hours off, with no more than normal maintenance and replacement of parts. Compliance shall be determined by examination of the preproduction sample. (See 3.2.)

(b) *Temperature.* — Any ambient temperature in the range of —40° to +150°F. (The 150°F. temperature includes effect of sun-load.) (See 4.12.1.)

(c) *Elevation.* — Any elevation up to 10,000 feet above sea level.

(d) *Relative humidity.* — Any relative humidity up to 98 percent.

3.23.2 *Equipment nonoperating.* — The equipment shall meet the requirements of 3.23.1 after subjection to any one or more of the following nonoperating conditions:

(a) *Temperature.* — Continuous exposure for 72 hours at 160°F. and 72 hours at —80°F., at any orientation.

(b) *Elevation.* — Any elevation up to 25,000 feet above sea level.



Source: https://assist.dla.mil -- Downloaded: 20261O-17T23:53Z
Check the source to verify that this is the current version before use.

(c) *Relative humidity.* — Relative humidity up to 100 percent, including condensation caused by temperature changes. (See 4.12.2.) Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.23.1 except that the output may be 25 percent less than specified value.

(d) *Rain.* — Two inches of rainfall per hour and wind at 25 miles per hour. (See 4.12.3.)

(e) *Orientation.* — Indefinite storage in any orientation.

(f) *Vibration, bounce, and shock; drop.* — See 3.24.

3.24 **Vibration, bounce, and shock: drop.**

3.24.1 *Vibration.* — Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second.

3.24.2 *Bounce and shock: drop.* — The equipment shall meet the following requirements:

| *Requirement* | *Inspection paragraph* | *Performance after inspection* |
|---|---|---|
| Bounce, light equipment | 4.12.5 | Full specification performance. No physical damage except surface abrasion. |
| Shock; drop | 4.12.6 | Equipment shall be operable. Minor mechanical damage is permissible. |

3.25 **Technical literature, tools, and running spare parts.**—Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.26 **Circuit diagrams.**—Circuit diagrams (schematic and wiring) shall be furnished as required to facilitate installation, operation, and maintenance of the equipment. The diagrams shall be in the form of labels meeting the general requirements of MIL-P-11268.

3.26.1 *Schematic and wiring diagrams.*— Schematic and wiring diagrams shall be suitable for use by maintenance personnel, and shall be mounted so far as practicable on inside surfaces where they will not be exposed to abrasion or the elements but will be readily accessible to maintenance personnel. Graphical symbols on schematic diagrams shall conform to Standard MIL-STD-15. Schematic and wiring diagrams shall conform to the sample drawings shown in Table I.

TABLE I. *Sample schematic and wiring diagrams*

| Type of diagram | Drawing number |
|---|---|
| Schematic diagram of electronic component | SC-D-15800 |
| Wiring diagram of an electrical component | SC-D-15801 |
| Schematic and wiring diagram of an electronic component | SC-D-20096 |

3.27 **Identification of equipment.** — The equipment shall be marked for identification in accordance with Standard MIL-STD-130.

3.27.1 *Serial numbers.* — Reproducing Equipment MC-364-( ) and Public Address Set AN/UIH-2( ) shall be serial numbered.

3.28 **Workmanship.**—The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following:

In Specification MIL-P-11268:

General requirements for plastic material

Wiring and cabling including:

Slack
Protection
Clearance
Splicing
Grounding, general



Source: https://assist.dla.mil -- Downloaded: 202610-17T23:53Z
Check the source to verify that this is the currentversion before use.

Grounding to chassis
Shielding on wire and cable
Flux and cleaning agents for soldering
Process for soldering
Cleaning of equipment
Finish of mounted hardware
Riveting
General requirements for securing of parts
Cleaning of welding
Process for welding
Self-tapping screws.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection, general.—Government inspection shall consist of the inspection specified in 4.1.1 thru 4.1.3.1.

4.1.1 *Classification of inspection.* — Inspection shall be classified as follows:

(a) Preproduction inspection
(b) Acceptance inspection

4.1.2 *Preproduction inspection.* — Preproduction inspection shall consist of the group A inspection and the preproduction inspection specified in Table II. Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling and (4) shock, ballistic or drop; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE II. *Preproduction inspection*

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Temperature | 3.23 | 4.12.1 |
| Moisture-resistance | 3.23 | 4.12.2 |
| Rain | 3.23 | 4.12.3 |
| Vibration | 3.24.1 | 4.12.4 |
| Bounce | 3.24.2 | 4.12.5 |
| Shock: drop | 3.24.2 | 4.12.6 |
| Suppression | 3.20 | 4.12.7 |

4.1.3 *Acceptance inspection.* — Acceptance inspection shall consist of group A, inspection as specified in 4.1.3.1 through 4.1.4. Each unit which will be subjected to Group A inspection shall be preconditioned in accordance with 4.1.5.

4.1.3.1 *Group A inspection.* — Group A inspection (including sampling) shall conform to Table III and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:53Z
Check the source to verify that this is the current version before use.

TABLE III. *Government group A inspection*

| Inspection | Requirement paragraph | Test paragraph | AQL |
|---|---|---|---|
| Visual and Mechanical | | | |
| *Cable Assemblies | 3.8, 3.9, 3.10 | MIL–C–3885 | |
| Microphone M-23( )/U, major | 3.7 | 4.2 | 1% |
| Microphone M-23( )/U, minor | | | 4% |
| Microphone M-23( )/U, control | | | 10% |
| RP-104( )UIH-2 or MX-932( )/U, major | 3.11, 3.12 | | 4.0 DPHU** |
| RP-104( )UIH-2 or MX-932( )/U, minor | | | 15 DPHU** |
| RP-104( )UIH-2 or MX-932( )/U, control | | | 10% |
| Dimensions | | 4.10 | 0.65% *** |
| Electrical | | | |
| *Cable Assemblies | 3.8, 3.9, 3.10 | MIL–C–3885 | |
| Microphone M-23( )/U | 3.7 | 4.13 | |
| Reproducer RP-104( )/UIH-2 or Turntable MX-932( )/U | | | |
| Operating voltages | 3.13 | 4.3 | 2.5 DPHU** combined |
| Power consumption | 3.14 | 4.4 | |
| Power output and distortion | 3.15 | 4.5 | |
| Frequency response | 3.16 | 4.6 | |
| Hum level | 3.17 | 4.7 | |
| Needle pressure and turntable speed | 3.18, 3.19 | 4.8, 4.9 | 1% |
| Operational | | 4.11 | 1% |

* Tests shall apply for each cable assembly individually.
** Defects per hundred units.
*** All electrical defects are considered major.

4.1.4 *Disposition of nonconforming product.* — When defective sample units or rejected lots are resubmitted for acceptance, such items shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the items are resubmitted and shall become the property of the Government.

4.1.5 *Bounce preconditioning.*—After final assembly, each component which will be examined, measured, or tested shall be preconditioned as follows: The component shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The component shall be subjected to this preconditioning for 1 minute.

4.2 *Visual and mechanical inspection.*— Parts and equipment shall be examined for workmanship, mechanical fit, loose nuts and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be examined for mechanical operation. Wiring, soldered connections, ground connections, welds, finishes, etc., shall be examined for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of Standard MIL–STD–105.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:53Z
Check the source to verify that this is the current version before use.

4.3 Operating voltages.—Operating voltages of the equipment shall be tested for compliance with the requirements of 3.13.

4.4 Power consumption.—The power consumption of the complete equipment shall be tested for compliance with the requirements of 3.14.

4.5 Power output and distortion.—The power output and distortion of the amplifier shall be measured for conformance with the requirements of 3.15 as follows: All voltage measurements shall be made using a high impedance vacuum tube voltmeter.

(1) Turn tone controls full on and phono gain off.
(2) Apply a .0015 volt, 1000 cycle signal from a calibrated audio oscillator to the microphone input channel.
(3) Adjust the microphone gain control until an output of 5.5 volts is obtained across a 3 ohm noninductive load resistor.
(4) Measure the total harmonic distortion present in the output signal by means of a properly calibrated distortion analyzer.
(5) The distortion shall be not more than 3 percent.
(6) Repeat steps (1) through (4) but with the microphone gain adjusted to obtain a 6 volt input.
(7) The distortion shall be not more than 5 percent.
(8) Turn the tone control full on and the microphone gain off and apply a 0.1 volt, 1000 cycles signal to the phono channel. Adust the phono gain for an output of 5.5 volts. Repeat steps (1) through (4).
(9) The distortion shall be not more than 3 percent.
(10) Repeat (8) but with the phono gain adjusted for 6 volt output.
(11) The distortion shall be not more than 5 percent.

4.6 Frequency response.—The frequency of the amplifier shall be measured for conformance with the requirements of 3.16 as follows:

(1) Repeat steps (1) to (3) of 4.5.
(2) Keep the signal input at 0.0015 volts, vary the oscillator over the range of 70 to 6000 cycles.
(3) Record output voltages in db with reference to output voltage at 1000 cycles.
(4) Values shall fall within the limits of figure 1.

4.7 Hum level.—The hum level of the reproducing equipment shall be measured for compliance with the requirements of 3.17.1 and 3.17.2 by first measuring the "no signal" output across the 3 ohm load resistor, with the microphone gain and tone controls turned full on the phono gain off, and secondly with the phono gain and tone control turned full on and the microphone gain off. Measurements shall be taken with the amplifier installed in its case.

4.8 Turntable speeds. — The turntable speeds of the equipment shall be tested for compliance with the requirements of 3.18.1 or 3.18.2.

4.9 Needle pressure.—The needle pressure of the pickup shall be tested for compliance with the requirements of 3.19.

4.10 Dimensional inspection.—Selected dimensions shall be gaged or measured to determine compliance with this specification. The dimensions to be checked shall be selected at a meeting between the contractor and the contracting officer, held as soon as practical after the manufacturers shop drawings have been prepared, but before equipment is selected for acceptance.

4.11 Operational test. — The assembled equipment shall be given a complete operational test of reproducing and public address use to determine that all circuits function properly and that overall performance is satisfactory.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:53Z
Check the source to verify that this is the current version before use.

4.12 Service conditions tests.—The following service conditions tests will be performed on the preproduction samples at a Government laboratory. Between tests, the equipment will be checked for such defects as loss of electrical continuity, loose parts, breakage, etc. Following the completion of the service conditions tests, the performance of the equipment will be checked and comparison of performance characteristics will be made with the initial performance tests results. Without other than normal adjustment, the operation of the equipment following the service conditions tests shall meet the requirements of 4.3 to 4.11 inclusive.

4.12.1 *Temperature.*—The equipment shall be subjected to the temperature cycle shown on Drawing SC–D–19999. The tests of 4.3 to 4.11 inclusive shall be made at step 3 and 8. The equipment shall conform to full specification requirements.

4.12.2 *Moisture-resistance test for equipment.*

4.12.2.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.12.2.2 *Test procedure.* — The equipment will be tested as follows:

(a) Dry at 130° ±5° F. for 24 hours.

(b) Condition at 77° ±5° F. and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.12.2.3 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Drawing SC–D–15914. The measurements specified in 4.12.2.3 shall be made during the periods shown on the drawing and shall comply with 4.12.2.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5° F. and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the components. No repair or replacement or parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.12.2.3.

4.12.2.3 *Performance.* — Measurements made during test, and specified performance during cycling, shall not exceed 10 percent of the measurements specified in 4.5, 4.7 and 4.11.

4.12.2.4 *Failure.* — If any component fails to meet the performance specified in 4.12.2.3 during cycling, it shall be realined or readjusted once. If the component then fails to meet 4.12.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the component fails to meet full specification requirements after conditioning and adjustment, as specified in 4.12.2.2.(e), it does not pass the test.

4.12.3 *Rain test.* — The equipment shall be tested, step by step, as follows:

4.12.3.1 Dry at 150° ± 5° F. for 48 hours.

4.12.3.2 Condition for 4 hours at 77° ± 5° F. and 40 to 60 percent relative humidity.

4.12.3.3 Take the measurements specified in 4.5 and 4.6 to insure that the equipment meets specified performance.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:53Z
Check the source to verify that this is the current version before use.

4.12.3.4 Place the closed equipment in its normal operation position, in a test chamber maintained at 77° ±5° F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ±5° F. and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

4.12.3.5 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for 1 hour.

4.12.3.6 Rotate the equipment 90° horizontally, to expose a second side of the equipment to the wind and rain. Maintain this condition for 1 hour.

4.12.3.7 Continue as in 4.12.3.6, until all four sides of the equipment have been exposed to the wind and rain, each side for 1 hour.

4.12.3.8 Open the equipment. There shall be no visible evidence of leakage.

4.12.3.9 Immediately after completion of the foregoing, repeat the measurements specified in 4.12.3.3 without removal from the chamber but with the rain and wind discontinued. Complete the measurements within 1 hour. The measurements shall meet specified performance.

4.12.4 *Vibration test.*—The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.24.1, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. The vibration table shall provide approximately sinusoidal vibration. (See 4.12.4.(d).)

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at $\frac{1}{64}$ inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.12.5 *Bounce test.*—The equipment, in its transit case, shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 r.p.m. ± 1 percent, for a total of 8 hours. During each ½-hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.12.6 *Shock test; drop.*—The equipment, as packaged for shipment, shall be dropped a distance of 6 inches on each corner, edge, and



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:53Z
Check the source to verify that this is the current version before use.

face (total of 26 drops). Drops shall be made from a quick-release hook; an Acme Drop tester, or equal; a Conbur Tester (10° inclined plane), or equal; or a basket-type pendulum tester. The floor or barrier receiving the impact shall be 2-inch fir, backed by concrete or a rigid steel frame. A mat of sponge rubber one inch thick shall be interposed as a damping medium between the equipment and the platform.

4.12.7 *Suppression.*—The equipment shall be tested for compliance with the requirements of 3.21.

4.13 **Microphone M–23( )/U.** — Microphone M-23 shall be tested for compliance with the requirements of paragraph 3.7.

## 5. PREPARATION FOR DELIVERY

### 5.1 Preservation and Packaging.

5.1.1 *For immediate use at first domestic receiving activity.*—Public Address Set (Reproducing Equipment MC–364–( )) and Public Address Set AN/UIH–2( ), shall be preserved and packaged to afford protection against deterioration and damage during shipment.

5.1.2 *For entry into the supply system.*—PRESERVE and package Public Address Set (Reproducing Equipment MC–364–( )) and Public Address Set AN/UIH–2( ), in accordance with procedures specified for designated methods as prescribed in Specification MIL–P–116 and further described in the following paragraphs:

5.1.2.1 *Technical literature.*—Package each technical literature Method IC–3 as follows: Inclose each literature within a close-fitting, Class "b" bag, conforming to Specification JAN–P–117. Properly seal closure.

5.1.2.2 *Tubes.*—Remove all operating tubes over 4 inches in length not securely clamped in place and package together with spare tubes, determined in quantity and number from bid request or contract, individually in accordance with the applicable provisions of Specification MIL–P–75.

5.1.2.3 *Electrical spare parts items.* — Package each electrical spare parts item, determined in quantity and number from bid request or contract, in accordance with the applicable methods described in Specification MIL–P–116.

5.1.2.4 *Accessories (electrical).*—Package each accessory in accordance with the applicable method described in Specification MIL–P–116.

5.1.2.5 *Speaker Assembly LS–148( )/U.* —Package in accordance with Specification MIL–L–13078.

5.1.2.6 *Turn Table MX–932( )/U and Reproducer RP 104( )/UIH–2.*—Package each turntable and reproducer individually Method IIb, as follows: Stow and secure spare parts and accessories packaged as described in 5.1.2.2 through 5.1.2.4 within designated spaces of the turntable and reproducer cases. Lock reproducer arm and turntable unit within case in place with the locking devices provided. Position the turntable and reproducer case cover over each unit and secure fastenings. Cushion each item on all surfaces with cells or pads or both fabricated of corrugated fiberboard conforming to Specification 19–10 and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned item together with the required amount of desiccant as prescribed in Specification MIL–P–116, within a close-fitting suitable style and grade corrugated fiberboard box. Seal entire closure with gummed paper tape conforming to Specification UU–T–111. Blunt all corners of the box. Place the boxed item within a water-vapor-proof barrier, qualified for Class 1 application in accordance with the requirements of Specification MIL–B–131. Heat seal closure. Place the water-vapor-proofed item within a close-fitting



Source: https://assist.dla.mil -- Downloaded: 20260-17T23:53Z
Check the source to verify that this is the current version before use.

suitable style and grade fiber board box conforming to Specification JAN-P-108. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape, conforming to Specification JAN-P-127.

5.2 **Packing.**

5.2.1 *For immediate use at first domestic receiving activity.*—Public Address Set (Reproducing Equipment MC-364-( )) and Public Address Set AN/UIH-2( ), shall be so packed as to insure arrival at destination in satisfactory condition and in containers complying with Uniform Freight Classification Ratings Rules, and Regulations.

5.2.2 *For domestic shipment and indefinite storage.*—Place one each Speaker Assembly LS-148 and one each Turntable MX-932 or one each Speaker Assembly LS-148 and one each Reproducer RP-104, packaged as specified in 5.1.2.5 and 5.1.2.6 together within a wood cleated plywood or nailed wood box conforming to the requirements of Specification PPP-B-601, or NN-B-621. Fabricate the box to fit the contents snugly. Place the technical literature packaged as specified in 5.1.2.1 between the packaged items and lid of the shipping container. Secure with water-resistant tape. Close the shipping container as specified in the appendix of the applicable box specification.

5.2.3 *For overseas shipment.*—Place one each Speaker Assembly LS-148 and one each Turntable MX-932 or one each Speaker Assembly LS-148 and one each Reproducer RP-104, packaged as specified in 5.1.2.5 and 5.1.2.6, together within a wood cleated plywood or nailed wood box conforming to the requirements of Specification PPP-B-601 or JAN-P-106. Fabricate the box to fit the contents snugly. Place the technical literature packaged as specified in 5.1.2.1 between the packaged items and lid of the shipping container. Secure with water resistant tape. Close the shipping containers as specified in the appendix of the applicable box specifications.

5.2.3.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification *only* for direct shipment to ports.

5.3 **Marking for shipment.**—Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of Standard MIL-STD-129.

6. **NOTES**

6.1 **Intended use.**—Reproducing Equipment MC-364-( ) and Public Address Set AN/UIH-2( ) are self-contained portable equipments used to entertain or address troops, indoors or out, with programs received by radio, reproduced from phonograph records or transcriptions, or picked up locally by microphone.

6.2 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required.
(c) Type packaging and packing required for shipment: (Immediate use at first domestic receiving activity; domestic shipment and indefinite storage; overseas shipment.)
(d) Samples required. (See 3.2.1.)
(e) Marking and shipping of samples.
(f) Place of final inspection.
(g) Facsimiles required.
(h) Technical literature required. (See 3.25.)
(i) Quantity of tools and running spare parts required. (See 3.25.)

6.3 **Nomenclature.**—The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: MC-364-W or AN/UIH-2W. As soon as possible after the award of the contract, the contractor should apply to the contracting officer for such information. (See 1.1.)



Source: https://assist.dla.mil -- Downloaded: 202640-17T23:53Z
Check the source to verify that this is the currentversion before use.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Air Force



Source: https://assist.dla.mil -- Downloaded: 202610-17T23:53Z
Check the source to verify that this is the current version before use.

# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4263, WHICH IS OBSOLETE. PLATE NO. 15419

Source: https://assist.dla.mil - Downloaded: 2026-05-14T23:53:53Z
Check the source to verify that this is the current version before use.